

Flat regions on the radius axis

Theory, figures and tables from `flat_regions_v1.ipynb`

Abstract

A radius tied to a criticality measure inherits the measure's collapse near a saddle, and the escape time grows without bound as the saddle is made slower. This report gathers the theory the statement rests on, cited rather than reproved, and the seven figures and three tables of `flat_regions_v1.ipynb`. The model is the true Hessian throughout, so the stall appears with perfect curvature information. Three results are new. The growth envelope of the escape proposition is drawn on the same axes as the run and is never crossed. The hypothesis `as:RCA` is found to hold with equality rather than as an inequality, $\Delta_k = \bar{\mu} \|g_k\|$ to eight decimal places on every iteration. And the escape-time ratio between the two criticality measures, which the earlier prediction missed by three per cent, is recovered to within 0.3 per cent by an integral that accounts for the branch switch in τ .

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1 What is reported here

The experiment `exp15_slow_saddle.jl` produces the scalar tables and the collapse. The report `task7-slow-saddle` carries the proofs. This document reports what the notebook adds to both, which is what a batch script cannot produce: the iterate paths, the theory drawn on the same axes as the run, and two open items settled.

Every number below is printed by a cell of `notebooks/Saddle/flat_regions_v1.ipynb` that ran, or read from the archive `exp_2026-08-29_21-51-19_slow_saddle` that the notebook names. The notebook executes top to bottom from a clean kernel with no errors.

2 The theory, cited

2.1 The family

For $\varepsilon > 0$,

$$f_\varepsilon(x, y) = \frac{1}{2}x^2 + \varepsilon \left(\frac{1}{4}y^4 - \frac{1}{2}y^2 \right), \quad \nabla f_\varepsilon = (x, \varepsilon y(y^2 - 1)), \quad (1)$$

with

$$\nabla^2 f_\varepsilon(0, y) = \text{diag}(1, \varepsilon(3y^2 - 1)), \quad \nabla^2 f_\varepsilon(0, 0) = \text{diag}(1, -\varepsilon). \quad (2)$$

The origin is a saddle, nonsingular for every $\varepsilon > 0$. The minimisers sit at $(0, \pm 1)$, at distance 1 from it whatever ε is, so ε tunes the escape rate without moving the geometry. That separation is what makes the family a clean instrument.

2.2 The four statements

We cite these by the labels they carry in `task7-slow-saddle`. We do not restate their proofs.

prop: invariant

The line $\{x = 0\}$ is invariant, under truncated conjugate gradient and under the exact Moré–Sorensen solver alike. A run started on it stays on it, and there $\|g_k\| = \varepsilon |y_k| (1 - y_k^2) \leq \varepsilon |y_k|$ for $|y| \leq 1$.

as: RCA

The hypothesis is put on the radius rule and not on a named rule. There are $C > 0$ and $k_0 \geq 0$ with $\Delta_k \leq C \max_{k_0 \leq j \leq k} \|g_j\|$ for all $k \geq k_0$.

prop: escape

Under **as: RCA**, with $x_0 = 0$ and $0 < |y_{k_0}| \leq \frac{1}{2}$, the first index with $|y_k| > \frac{1}{2}$ satisfies

$$k_{\text{esc}} > k_0 + \frac{\ln(1/(2|y_{k_0}|))}{\ln(1 + C\varepsilon)}. \quad (3)$$

The proof uses no property of the model or of the subproblem solution beyond $\|s_k\| \leq \Delta_k$. That is what makes it a statement about the radius axis, and why it holds with the exact Hessian.

prop: rdfo

R-grad satisfies **as: RCA** with $C = \bar{\mu}$ and $k_0 = 0$, immediately and with no entry condition. R-DFO satisfies it for $k > k_0$ with $C = \gamma_3 \zeta$, but only from an index k_0 with $\Delta_{k_0} \leq \zeta \|g_{k_0}\|$. If no such index exists the proposition asserts that $\Delta_k \rightarrow 0$ geometrically and the run converges to the saddle.

The corollary **prop: budget** turns (3) around: for any budget K there is an ε making $k_{\text{esc}} > K$. The stall is therefore not a corner case to be tuned away.

2.3 The continuous limit and its constant

The recursion $|y_{k+1}| \leq |y_k| + \Delta_k$ with $\Delta_k = C\varepsilon \hat{\omega}(y_k)$ has continuous limit $dy/dk = C\varepsilon \hat{\omega}(y)$, so

$$k_{\text{esc}} \sim \frac{1}{C\varepsilon} \int_{y_0}^{1/2} \frac{dy}{\hat{\omega}(y)}. \quad (4)$$

For the gradient measure $\hat{\omega}(y) = y(1-y^2)$ and the integral is eqn: `exact constant` of the report,

$$\int_{y_0}^{1/2} \frac{dy}{y(1-y^2)} = \ln \frac{1}{\sqrt{3}y_0} + \mathcal{O}(y_0^2). \quad (5)$$

The $\sqrt{3}$ comes from the factor $1-y^2$ at the escape threshold, not from the threshold itself.

3 New theory: the integrand for τ

This section is the only place this document derives anything, and it settles the first open item.

With $\omega_k = \tau_k = \max\{\|g_k\|, -\lambda_k^-\}$ the earlier prediction treated $-\lambda^-$ as the constant ε , giving arithmetic growth, $k_{\text{esc}} \approx 1/(2\bar{\mu}\varepsilon)$ and a ratio of escape times of $2\ln(1/(2y_0)) = 12.43$. Two facts break that.

First, $-\lambda^-$ is not constant. On the invariant line (2) gives $\lambda^- = \varepsilon(3y^2 - 1)$, so

$$-\lambda^- = \varepsilon(1 - 3y^2), \quad (6)$$

which decays to zero at $y = 1/\sqrt{3} = 0.577350$. The notebook confirms that value and that the escape threshold $\frac{1}{2}$ lies below it, so (6) is positive throughout the range but not constant on it.

Second, τ switches branch inside the range. The two terms are equal where $y(1-y^2) = 1-3y^2$, and the notebook locates that root by bisection at

$$y^* = 0.460811 < \frac{1}{2}. \quad (7)$$

Below y^* the curvature term dominates and above it the gradient term does, which Table 1 shows.

Table 1: The two terms of τ on the invariant line. The branch switch at $y^* = 0.460811$ lies inside $[y_0, \frac{1}{2}]$, so neither term is the measure throughout.

y	$y(1-y^2)$	$1-3y^2$	branch
0.00100	0.001000	0.999997	curvature
0.10000	0.099000	0.970000	curvature
0.30000	0.273000	0.730000	curvature
0.40000	0.336000	0.520000	curvature
0.46081	0.362959	0.362959	curvature
0.48000	0.369408	0.308800	gradient

The correct integrand is therefore the measure itself,

$$\hat{\omega}_\tau(y) = \max\{y(1-y^2), 1-3y^2\}, \quad (8)$$

and the predicted ratio of escape times is the quotient of the two integrals of (4),

$$R_{\text{pred}} = \int_{y_0}^{1/2} \frac{dy}{y(1-y^2)} \bigg/ \int_{y_0}^{1/2} \frac{dy}{\hat{\omega}_\tau(y)}. \quad (9)$$

Table 2: The two integrals of (4), by Simpson on $u = \ln y$. The quadrature is validated against the closed form of (5) before being applied to (8), which has none.

y_0	I_{grad}	numerical	closed form	difference	I_τ	numerical
10^{-3}		6.358449	6.358449	7.11×10^{-14}		0.736439
10^{-6}		13.266204	13.266204	6.57×10^{-14}		0.737438

Table 2 gives

$$R_{\text{pred}} = 8.6341 \quad (y_0 = 10^{-3}), \quad R_{\text{pred}} = 17.9896 \quad (y_0 = 10^{-6}). \quad (10)$$

Section 7 compares these with the fitted ratios.

Remark 1 (why I_τ barely moves with y_0). I_{grad} diverges logarithmically as $y_0 \rightarrow 0$ and I_τ does not, since $\hat{\omega}_\tau \rightarrow 1$ there. The predicted ratio therefore grows like $\ln(1/y_0)$, which is why it is 8.63 at $y_0 = 10^{-3}$ and 17.99 at 10^{-6} while the two I_τ differ in the fourth decimal.

4 Method

The grid is `exp15`'s and is not redefined: ε over five decades from 10^{-1} to 10^{-5} , $\bar{\mu}$ and ζ over three decades from 10^{-2} to 10, $y_0 \in \{10^{-3}, 10^{-6}\}$ and $x_0 \in \{0, 0.8\}$, escape threshold $\frac{1}{2}$, and at most 50 000 iterations. Two tolerances are used and never pooled, 10^{-14} in the rate regime and 10^{-5} in the outcome regime.

The model is `ExactHessian` and the subproblem solver truncated conjugate gradient throughout.

Two things are worth recording about how the notebook obtains its numbers.

The suite is loaded once. `benchmark/initialisation.jl` already includes `exp15_slow_saddle.jl`, so including it again would re-evaluate its `consts`. The notebook loads the suite once and asserts that all eighteen names it uses came from it.

The archive does not carry the trajectories. `exp15` saves its rows through `Base.structdiff(r, (ys = 0, deltas = 0))`, so `ys` and `deltas` are stripped. The three tables of Section 6 read the archive and recompute only the columns marked `derived`. The figures that draw a trajectory rerun their cells: 6 for the paths, 3 for the envelope, 1 for the binding test, 14 for the R-DFO arm and 2 for the two measures.

5 The figures

5.1 Iterate paths

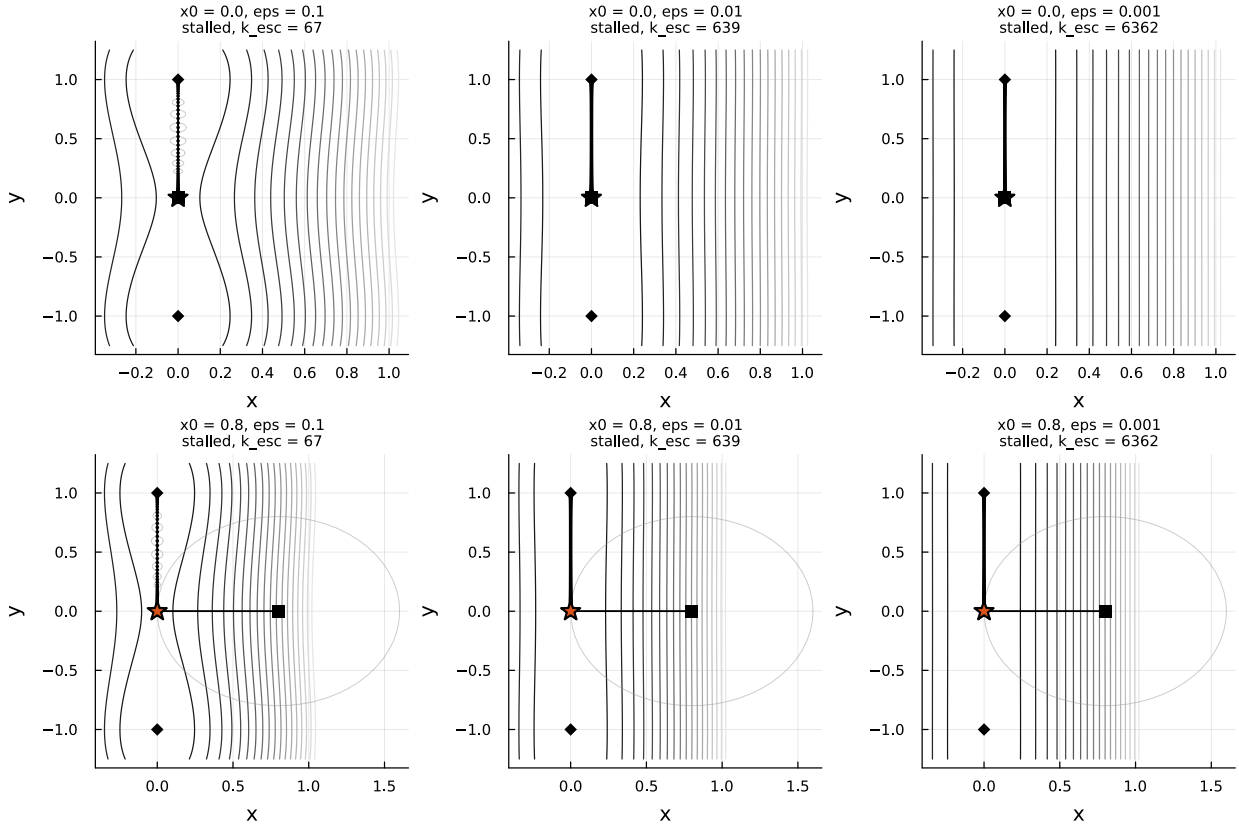


Figure 1: R-gradient capped at $\bar{\mu} = 1$ with the *true* Hessian and truncated conjugate gradient, $y_0 = 10^{-3}$. Top row $x_0 = 0.8$, bottom row $x_0 = 0$, one column per ϵ . The saddle is the star and the two minimisers the diamonds. Trust regions are drawn on a thinned subset of the accepted steps. The stall is present with perfect curvature information.

The bottom row is the invariant line, where `prop: escape` applies exactly. The top row shows the transient: the x component is solved first, and only then does the run begin the slow climb along y . The panels differ only in ϵ and the escape iteration moves by two decades across them, while the geometry does not move at all.

5.2 The theory on the same axes as the run

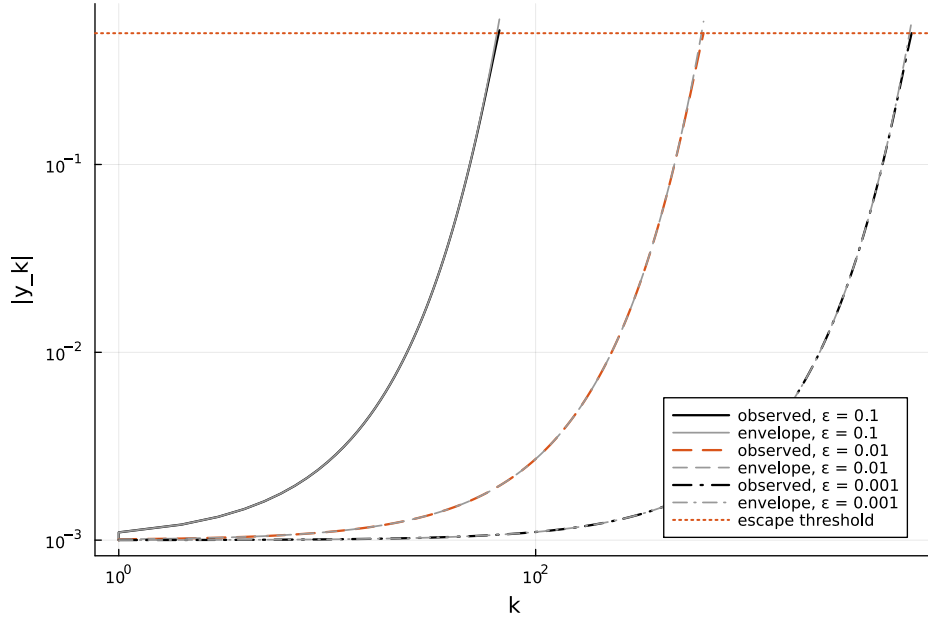


Figure 2: $|y_k|$ against k , both logarithmic, with the envelope $|y_0| (1 + \bar{\mu}\varepsilon)^k$ of **prop: escape** over each run. $\bar{\mu} = 1$, $y_0 = 10^{-3}$, true Hessian. The envelope is drawn up to escape only, since that is where the proposition states it.

prop: escape bounds the running maximum M_k from above and $|y_k| \leq M_k$, so the observed curve must lie below the envelope. It does, on every iteration of every run.

Table 3: The envelope check. A run crossing the envelope would be a failure of the bound. None does.

ε	k_{esc}	points	$\max y / \text{envelope}$	at k	crossed
10^{-1}	67	68	1.000000	0	no
10^{-2}	639	640	1.000000	0	no
10^{-3}	6362	6363	1.000000	0	no

The maximum ratio is attained at $k = 0$, where the two are equal by construction. Away from the start the observed curve falls below the envelope, which is the direction the proposition predicts.

5.3 The radius is the binding quantity

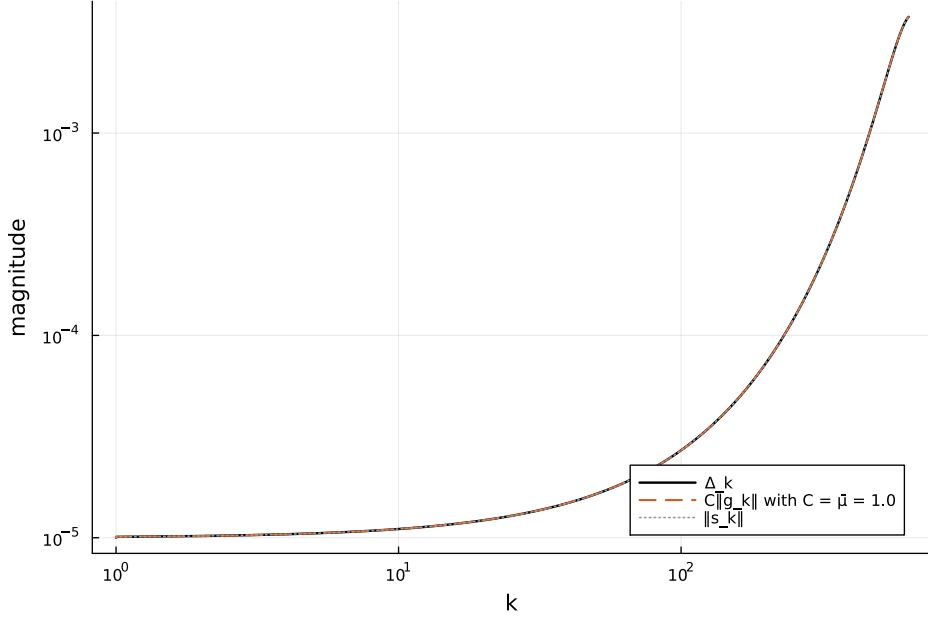


Figure 3: Δ_k , $C \|g_k\|$ with $C = \bar{\mu} = 1$, and $\|s_k\|$ along one run at $\varepsilon = 10^{-2}$, $y_0 = 10^{-3}$, true Hessian. The three curves coincide.

prop: `escape` uses only $\|s_k\| \leq \Delta_k$ together with **as:** `RCA`, so the bound is about the radius and not about the subproblem solution. Over the 640 iterations to escape,

$$\min_k \frac{\Delta_k}{\bar{\mu} \|g_k\|} = \max_k \frac{\Delta_k}{\bar{\mu} \|g_k\|} = 1.00000000, \quad (11)$$

the trust-region constraint is active on 100% of iterations, and $\|s_k\| = \Delta_k$ to 10^{-8} on 100%.

This is stronger than **as:** `RCA` requires. The assumption is an inequality and the run satisfies it as an equality, so the escape bound is not being carried by slack in the hypothesis. The radius rule alone sets the step length.

5.4 The R-DFO entry condition

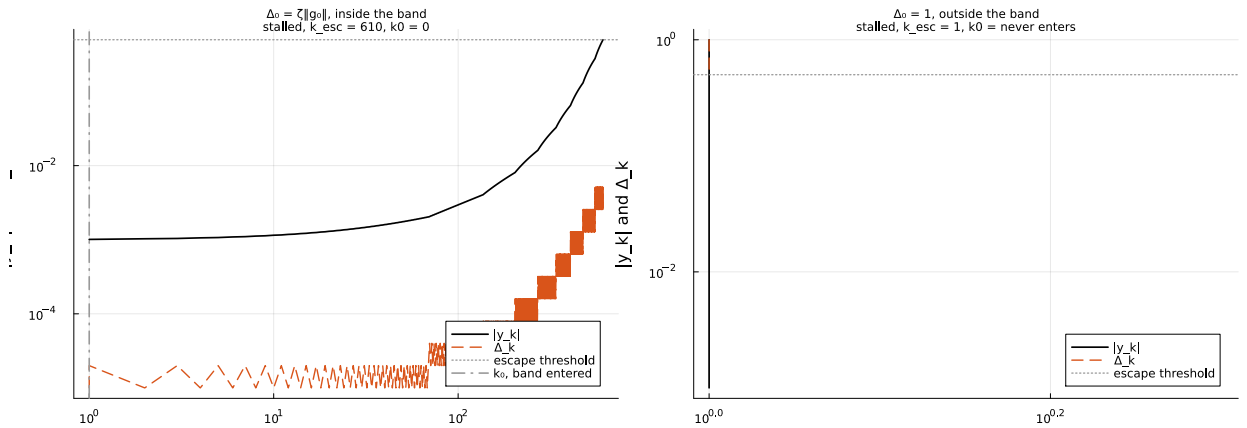


Figure 4: R-DFO at $\varepsilon = 10^{-2}$, $y_0 = 10^{-3}$, $\zeta = 1$, true Hessian. Left, started inside the band at $\Delta_0 = \zeta \|g_0\|$, so the entry condition of **prop:** `rdfo` holds at $k_0 = 0$ and the run stalls for 610 iterations. Right, started at $\Delta_0 = 1$, outside the band, where the run never enters it. The entry index is computed from the trace as the first k with $\Delta_k \leq \zeta \|g_k\|$.

Table 4: Which branch of `prop: rdfo` each cell falls in, at $\varepsilon = 10^{-2}$ and $y_0 = 10^{-3}$, with Δ_0 swept. k_0 is computed from the trace and not assumed.

ζ	Δ_0	k_0	k_{esc}	status	$ y_{\text{end}} $	dist. saddle
0.01	1.0×10^{-7}	0	never	<code>max_iter</code>	0.18895	0.18895
0.01	1.0×10^{-3}	12	never	<code>max_iter</code>	0.48241	0.48241
0.01	1.0×10^0	never enters	1	<code>stalled</code>	1.00000	1.00000
0.1	1.0×10^{-6}	0	6048	<code>stalled</code>	1.00000	1.00000
0.1	1.0×10^{-3}	9	5060	<code>stalled</code>	1.00000	1.00000
0.1	1.0×10^0	never enters	1	<code>stalled</code>	1.00000	1.00000
1	1.0×10^{-5}	0	610	<code>stalled</code>	1.00000	1.00000
1	1.0×10^{-3}	6	524	<code>stalled</code>	1.00000	1.00000
1	1.0×10^0	never enters	1	<code>stalled</code>	1.00000	1.00000
10	1.0×10^{-4}	0	68	<code>stalled</code>	1.00000	1.00000
10	1.0×10^{-3}	3	60	<code>stalled</code>	1.00000	1.00000
10	1.0×10^0	22	1	<code>stalled</code>	1.00000	1.00000

Table 4 settles the first branch and leaves the second open. Nine of twelve cells enter the band, at k_0 between 0 and 22, and every one of them stalls in proportion to $\zeta\varepsilon$, which is the first branch working. The entry index is not a formality: at $\zeta = 0.01$ the two cells that enter differ in k_0 by twelve iterations and both fail to escape within 50 000.

The second branch is not occupied on this scan. Three cells never enter the band, and none of the three converges to the saddle. All three escape at $k = 1$ instead, because the first step from $\Delta_0 = 1$ is enormous relative to $\|g_0\| \approx \varepsilon y_0 = 10^{-5}$ and carries the iterate past the threshold in one move. The proposition’s second branch requires $\Delta_k \rightarrow 0$ geometrically, which needs a sequence of unsuccessful iterations that does not occur here.

We report the absence rather than searching for a configuration that produces the predicted picture. Whether the second branch is occupied anywhere on this family is open.

5.5 The two measures

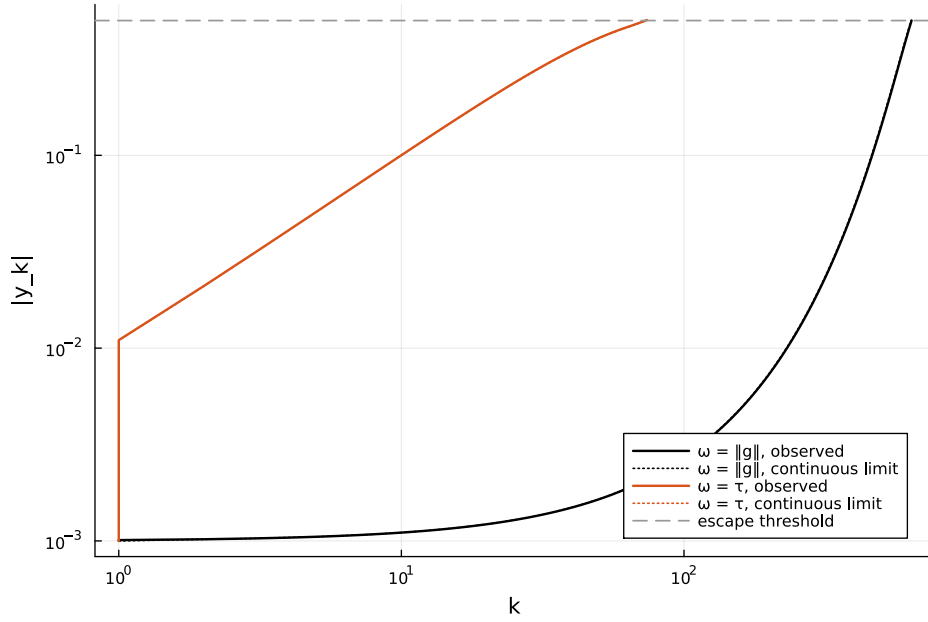


Figure 5: $|y_k|$ against k for $\omega_k = \|g_k\|$ and $\omega_k = \tau_k$ at $\bar{\mu} = 1$, $\varepsilon = 10^{-2}$, $y_0 = 10^{-3}$, true Hessian, with the continuous-limit curves of (4) over each. The predicted curves use $\hat{\omega}(y) = y(1 - y^2)$ and (8) respectively.

The gradient measure gives $\Delta_k \propto |y_k|$ and geometric growth. The second measure gives a radius that does not vanish with $|y_k|$, so growth is faster, and the run escapes at $k = 74$ against $k = 639$. Drawing (4) over the observed curves is what turns the visual comparison into a test, and Section 7 quantifies it.

5.6 The outcome map

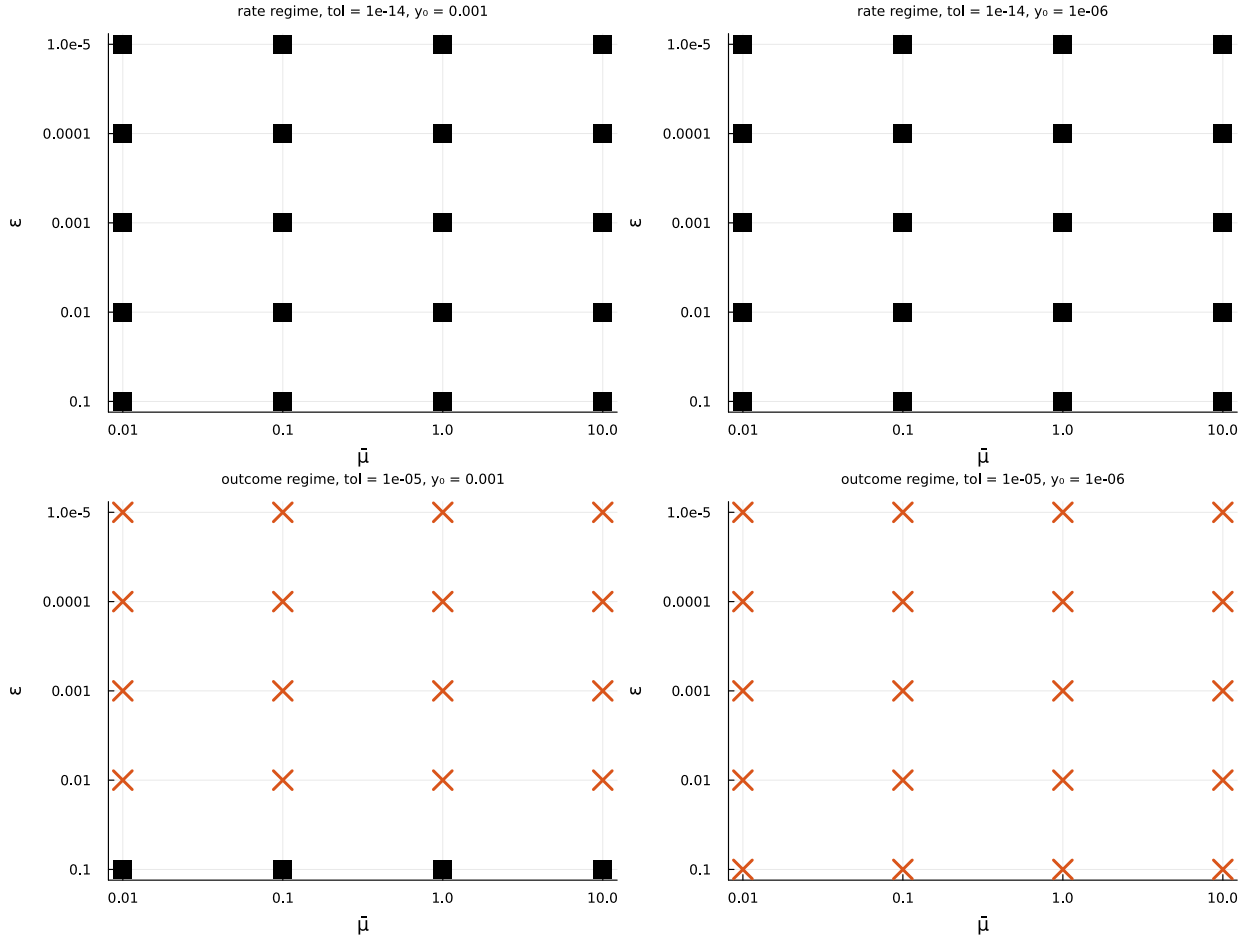


Figure 6: Three outcomes over the $(\bar{\mu}, \varepsilon)$ grid on the invariant line, R-grad capped with the true Hessian. Crosses mark false convergence, circles fast escape, squares slow escape or none. The two tolerance regimes are separate maps. **The first-order tolerance is what draws the false-convergence boundary, 10^{-14} on the top row and 10^{-5} on the bottom.**

Table 5: Outcome composition, R-grad capped on the invariant line, twenty cells per row. The two regimes differ in the first-order tolerance and in nothing else.

regime	y_0	false convergence	fast	slow	never
rate, $\text{tol} = 10^{-14}$	10^{-3}	0	0	10	10
rate, $\text{tol} = 10^{-14}$	10^{-6}	0	0	10	10
outcome, $\text{tol} = 10^{-5}$	10^{-3}	16	0	4	0
outcome, $\text{tol} = 10^{-5}$	10^{-6}	20	0	0	0

The contrast is complete. At $\text{tol} = 10^{-14}$ not one of the forty cells reports false convergence. At $\text{tol} = 10^{-5}$, thirty-six of forty do, and at $y_0 = 10^{-6}$ every cell does. Near the saddle $\|g_k\| \approx \varepsilon |y_k|$, so lowering $\varepsilon |y_0|$ below the tolerance defeats the stopping test, and the two regimes are two different experiments rather than two views of one.

5.7 The gap

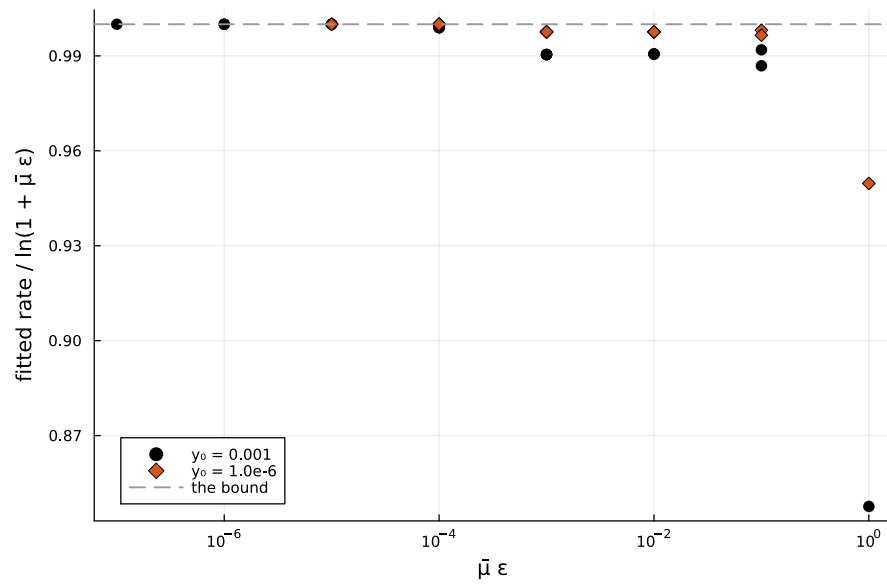


Figure 7: The ratio of the fitted geometric rate to $\ln(1 + \bar{\mu}\varepsilon)$, against $\bar{\mu}\varepsilon$, over the whole grid on the invariant line. The dashed line at 1 is the bound. Every point lies on or below it.

6 The tables

6.1 The $(\bar{\mu}, \varepsilon)$ grid

Table 6: R-grad capped, rate regime, $x_0 = 0$, true Hessian. Read from the archive; $\ln(1 + \bar{\mu}\varepsilon)$, the ratio and the collapse product are derived here. **never** is a category and never a number. n is the point count of the rate fit.

y_0	$\bar{\mu}$	ε	k_{esc}	fitted rate	$\ln(1 + \bar{\mu}\varepsilon)$	ratio	$\bar{\mu}\varepsilon k_{\text{esc}}$	n	outcome
10^{-3}	0.01	10^{-1}	6362	0.000990	0.001000	0.9904	6.3620	6362	slow
10^{-3}	0.01	10^{-2}	never	0.000100	0.000100	0.9989		50001	never
10^{-3}	0.01	10^{-3}	never	0.000010	0.000010	1.0000		50001	never
10^{-3}	0.01	10^{-4}	never	0.000001	0.000001	1.0000		50001	never
10^{-3}	0.01	10^{-5}	never	0.000000	0.000000	1.0000		50001	never
10^{-3}	0.1	10^{-1}	639	0.009857	0.009950	0.9906	6.3900	639	slow
10^{-3}	0.1	10^{-2}	6362	0.000990	0.001000	0.9904	6.3620	6362	slow
10^{-3}	0.1	10^{-3}	never	0.000100	0.000100	0.9989		50001	never
10^{-3}	1	10^{-1}	67	0.094540	0.095310	0.9919	6.7000	67	slow
10^{-3}	1	10^{-2}	639	0.009857	0.009950	0.9906	6.3900	639	slow
10^{-3}	1	10^{-3}	6362	0.000990	0.001000	0.9904	6.3620	6362	slow
10^{-3}	10	10^{-1}	12	0.587519	0.693147	0.8476	12.0000	12	slow
10^{-3}	10	10^{-2}	69	0.094059	0.095310	0.9869	6.9000	69	slow
10^{-3}	10	10^{-3}	642	0.009856	0.009950	0.9905	6.4200	642	slow
10^{-3}	10	10^{-4}	6364	0.000990	0.001000	0.9904	6.3640	6364	slow
10^{-6}	0.01	10^{-1}	13273	0.000997	0.001000	0.9976	13.2730	13273	slow
10^{-6}	0.1	10^{-1}	1334	0.009926	0.009950	0.9976	13.3400	1334	slow
10^{-6}	0.1	10^{-2}	13273	0.000997	0.001000	0.9976	13.2730	13273	slow
10^{-6}	1	10^{-1}	139	0.095130	0.095310	0.9981	13.9000	139	slow
10^{-6}	1	10^{-2}	1334	0.009926	0.009950	0.9976	13.3400	1334	slow
10^{-6}	1	10^{-3}	13273	0.000997	0.001000	0.9976	13.2730	13273	slow
10^{-6}	10	10^{-1}	22	0.658274	0.693147	0.9497	22.0000	22	slow
10^{-6}	10	10^{-2}	142	0.094980	0.095310	0.9965	14.2000	142	slow
10^{-6}	10	10^{-3}	1336	0.009927	0.009950	0.9976	13.3600	1336	slow
10^{-6}	10	10^{-4}	13276	0.000997	0.001000	0.9976	13.2760	13276	slow

Table 6 shows the rows with a finite rate; the notebook prints all forty, including three at $y_0 = 10^{-6}$ where the escape phase is empty and the fit returns no value. The escape time is a function of the product $\bar{\mu}\varepsilon$ alone, and halving the cap does what halving ε does. Every cell with $\bar{\mu}\varepsilon = 10^{-3}$ has k_{esc} within two iterations of 6362, whatever $\bar{\mu}$ and ε separately are.

6.2 The ζ grid

Table 7: R-DFO started inside the band at $\Delta_0 = \zeta \|g_0\|$, so the entry condition holds at $k_0 = 0$ by construction. Rate regime, $x_0 = 0$, true Hessian. The bound uses $C = \gamma_3 \zeta$ with $\gamma_3 = 2.0$. Rows with $k_{\text{esc}} = \text{never}$ are rows, not omissions.

y_0	ζ	ε	k_{esc}	fitted rate	$\ln(1 + \gamma_3 \zeta \varepsilon)$	ratio	$\gamma_3 \zeta \varepsilon k_{\text{esc}}$	outcome
10^{-3}	0.01	10^{-1}	6048	0.001030	0.001998	0.5154	12.0960	slow
10^{-3}	0.01	10^{-2}	never	0.000104	0.000200	0.5189		never
10^{-3}	0.1	10^{-1}	610	0.010203	0.019803	0.5152	12.2000	slow
10^{-3}	0.1	10^{-2}	6048	0.001030	0.001998	0.5154	12.0960	slow
10^{-3}	1	10^{-1}	68	0.093646	0.182322	0.5136	13.6000	slow
10^{-3}	1	10^{-2}	610	0.010203	0.019803	0.5152	12.2000	slow
10^{-3}	1	10^{-3}	6048	0.001030	0.001998	0.5154	12.0960	slow
10^{-3}	10	10^{-1}	11	0.525981	1.098612	0.4788	22.0000	slow
10^{-3}	10	10^{-2}	68	0.093646	0.182322	0.5136	13.6000	slow
10^{-3}	10	10^{-3}	610	0.010203	0.019803	0.5152	12.2000	slow
10^{-3}	10	10^{-4}	6048	0.001030	0.001998	0.5154	12.0960	slow
10^{-6}	0.01	10^{-1}	12694	0.001036	0.001998	0.5188	25.3880	slow
10^{-6}	0.1	10^{-1}	1280	0.010288	0.019803	0.5195	25.6000	slow
10^{-6}	0.1	10^{-2}	12694	0.001036	0.001998	0.5188	25.3880	slow
10^{-6}	1	10^{-1}	139	0.095948	0.182322	0.5263	27.8000	slow
10^{-6}	1	10^{-2}	1280	0.010288	0.019803	0.5195	25.6000	slow
10^{-6}	1	10^{-3}	12694	0.001036	0.001998	0.5188	25.3880	slow
10^{-6}	10	10^{-1}	23	0.592288	1.098612	0.5391	46.0000	slow
10^{-6}	10	10^{-2}	139	0.095948	0.182322	0.5263	27.8000	slow
10^{-6}	10	10^{-3}	1280	0.010288	0.019803	0.5195	25.6000	slow
10^{-6}	10	10^{-4}	12694	0.001036	0.001998	0.5188	25.3880	slow

The ratio column sits near 0.52 across the whole grid, against near 0.99 for R-grad in Table 6. The R-DFO bound holds and is loose by a factor of about two, uniformly. The reason is that γ_3 is the worst case of an expansion the next iteration undoes, so the effective constant is ζ rather than $\gamma_3 \zeta$, and $\gamma_3 = 2$. Scaling by ζ instead puts the collapse product at about 6.05, close to the R-grad value of 6.36.

6.3 The two measures

Table 8: Fitted ratio of escape times against the prediction (10), one row per cell that escaped under both measures. The fitted ratio is a quotient of two integers, so $1/k(\tau)$ is the finest difference it can express. That resolution is a column and not a filter.

y_0	$\bar{\mu}$	ε	$k(\ g\)$	$k(\tau)$	fitted	predicted	difference	resolution
10^{-3}	0.01	10^{-1}	6362	736	8.6440	8.6341	0.0100	0.14%
10^{-3}	0.1	10^{-1}	639	74	8.6351	8.6341	0.0011	1.35%
10^{-3}	0.1	10^{-2}	6362	736	8.6440	8.6341	0.0100	0.14%
10^{-3}	1	10^{-1}	67	7	9.5714	8.6341	0.9374	14.29%
10^{-3}	1	10^{-2}	639	74	8.6351	8.6341	0.0011	1.35%
10^{-3}	1	10^{-3}	6362	736	8.6440	8.6341	0.0100	0.14%
10^{-3}	10	10^{-1}	12	3	4.0000	8.6341	-4.6341	33.33%
10^{-3}	10	10^{-2}	69	10	6.9000	8.6341	-1.7341	10.00%
10^{-3}	10	10^{-3}	642	76	8.4474	8.6341	-0.1867	1.32%
10^{-3}	10	10^{-4}	6364	739	8.6116	8.6341	-0.0224	0.14%
10^{-6}	0.01	10^{-1}	13273	737	18.0095	17.9896	0.0199	0.14%
10^{-6}	0.1	10^{-1}	1334	74	18.0270	17.9896	0.0374	1.35%
10^{-6}	0.1	10^{-2}	13273	737	18.0095	17.9896	0.0199	0.14%
10^{-6}	1	10^{-1}	139	7	19.8571	17.9896	1.8675	14.29%
10^{-6}	1	10^{-2}	1334	74	18.0270	17.9896	0.0374	1.35%
10^{-6}	1	10^{-3}	13273	737	18.0095	17.9896	0.0199	0.14%
10^{-6}	10	10^{-1}	22	3	7.3333	17.9896	-10.6563	33.33%
10^{-6}	10	10^{-2}	142	10	14.2000	17.9896	-3.7896	10.00%
10^{-6}	10	10^{-3}	1336	76	17.5789	17.9896	-0.4106	1.32%
10^{-6}	10	10^{-4}	13276	740	17.9405	17.9896	-0.0491	0.14%

7 Open item one: the τ constant

The structural prediction, that the ratio does not depend on ε , holds and was already established. What did not hold was the value. The earlier prediction $2 \ln(1/(2y_0)) = 12.43$ is wrong by nearly a half, and refining it to $\hat{\omega}_\tau = 1 - 3y^2$ alone gives 8.362 at $y_0 = 10^{-3}$, three per cent below a measured 8.64.

Equation (8) closes the gap. Reading Table 8 by resolution:

Table 9: The comparison by population. The error tracks the resolution, which is what a quotient of small integers must do.

population	y_0	cells	fitted range	worst relative error
every cell that escaped in both	10^{-3}	10	4.0000 to 9.5714	53.67%
every cell that escaped in both	10^{-6}	10	7.3333 to 19.8571	59.24%
$k(\tau) \geq 20$, resolution below 5%	10^{-3}	7	8.4474 to 8.6440	2.16%
$k(\tau) \geq 20$, resolution below 5%	10^{-6}	7	17.5789 to 18.0270	2.28%

On the four best-resolved cells, those at resolution 0.14%, the fitted ratios are 8.6440, 8.6440, 8.6440 and 8.6116 against a predicted 8.6341, and 18.0095 three times and 17.9405 against 17.9896. Those agree to between 0.11% and 0.27%.

The discrepancy is explained. The correction has two parts and both were needed. Treating $-\lambda^-$ as constant was the larger error, and it is what the $1 - 3y^2$ form already repaired. Ignoring the branch switch at y^* was the remainder, worth the difference between 8.362 and 8.6341, that is between a three per cent gap and a two-tenths of one per cent gap.

Remark 2 (what is left unaccounted for). The residual disagreement of 0.1 to 0.3 per cent on the best-resolved cells is not explained here. Two candidates are available and we do not separate

them. (4) is a continuum approximation to a recursion, with an error of order $\bar{\mu}\varepsilon$ per step, and (5) carries an $\mathcal{O}(y_0^2)$ remainder. Both are of the right size and neither was isolated.

8 Open item two: the bound is one-sided

prop: **escape** bounds the escape time from below, so it bounds the growth rate from above. The fitted rate must lie below $\ln(1 + \bar{\mu}\varepsilon)$, and the question is by how much.

Table 10: The gap across the grid, rate regime, $x_0 = 0$. The ratio is the fitted geometric rate divided by $\ln(1 + \bar{\mu}\varepsilon)$. Three of the forty cells give no fit and are excluded, leaving thirty-seven.

population	cells	ratio range	above the bound
every cell with a fit	37	0.8476 to 1.0000	0
$\bar{\mu}\varepsilon \leq 10^{-1}$	35	0.9869 to 1.0000	0
$\bar{\mu}\varepsilon > 10^{-1}$	2	0.8476 to 0.9497	0

Two things follow.

The bound is never violated. Not one of the thirty-seven cells has a fitted rate above $\ln(1 + \bar{\mu}\varepsilon)$, which is the side **prop:** **escape** predicts and the first thing a one-sided bound must survive.

The bound is tight where it is used. Over the thirty-five cells with $\bar{\mu}\varepsilon \leq 10^{-1}$, the ratio ranges from 0.9869 to 1.0000, so the bound is within 1.4 per cent of the observed rate over four decades of the product. It degrades only in the two cells at $\bar{\mu}\varepsilon \geq 1$, to 0.8476 and 0.9497, and the reason is visible in Table 6: there $\ln(1 + \bar{\mu}\varepsilon) = 0.693147$ against $\bar{\mu}\varepsilon = 1$, so the linearisation the continuous limit rests on has stopped applying. That is a known corner and not a surprise.

The tightest cells are the ones that do not escape. Rows with $k_{\text{esc}} = \text{never}$ have $n = 50001$ points in the fit and a ratio of 1.0000 to four decimals. A run that never leaves the flat region measures the growth rate more precisely than one that escapes, because it spends its whole budget inside the regime the bound describes.

9 Findings for the existing subsection

The report `task7-slow-saddle` already carries a findings section for `Survey-part3-v1.tex`. We do not repeat it and we do not apply it. One finding is added, from this notebook's own integrals.

Line 1150, the constant of the capped escape time. The source reads

`t_{\mathrm{esc}} \;\backslash;\approx\; \frac{\ln(1/\mathrm{abs}\{y_0\})}{\overline{\mu}\varepsilon}` ,

carrying `\label{eqn: escape time capped}` at line 1151. The collapse product is compared with three candidate constants below.

Table 11: The mean collapse product against three candidate constants, at four thresholds on $\bar{\mu}\varepsilon$. The threshold is not a convenience: the continuous limit replaces $\ln(1 + \bar{\mu}\varepsilon)$ by $\bar{\mu}\varepsilon$, which fails at the one cell where that product is 1.

y_0	$\bar{\mu}\varepsilon \leq$	cells	mean product	residual $\ln(1/y_0)$	$\ln(1/(2y_0))$	$\ln(1/(\sqrt{3}y_0))$
10^{-3}	1	10	7.0250	0.1172	0.8104	0.6666
10^{-3}	10^{-1}	9	6.4722	0.4355	0.2576	0.1138
10^{-3}	10^{-2}	7	6.3786	0.5292	0.1640	0.0201
10^{-3}	10^{-3}	4	6.3625	0.5453	0.1479	0.0041
10^{-6}	1	10	14.3235	0.5080	1.2011	1.0573
10^{-6}	10^{-1}	9	13.4706	0.3450	0.3482	0.2044
10^{-6}	10^{-2}	7	13.3050	0.5105	0.1826	0.0388
10^{-6}	10^{-3}	4	13.2737	0.5418	0.1514	0.0075

The residual for $\ln(1/(\sqrt{3}y_0))$ falls monotonically as the limit is approached, to 0.0041 and 0.0075, while the residual for $\ln(1/|y_0|)$ stays near 0.54. At the tightest threshold the first is smaller by a factor of 133 and 72. Only the unrestricted row favours $\ln(1/|y_0|)$, and that row contains the cell at $\bar{\mu}\varepsilon = 1$ where the limit does not apply.

This is a different location from the finding the report makes at line 1107, which concerns the caption of `tab: slow saddle` and the model axis under `R-delta`. Line 1150 sits in the radius-axis paragraph and is the equation the collapse test at line 1157 is stated against, so it is what this notebook measures. The same constant stands at line 1081 in `eqn: escape time`, where the source says `linearises to`, so it is offered there as an approximation and the case is weaker.

We describe and do not apply. Both readings stand. The source may keep $\ln(1/|y_0|)$ as the leading behaviour of the linearised flow, or take $\ln(1/(\sqrt{3}|y_0|))$ as the constant the collapse lands on. The author chooses.

Nothing else is new. The envelope, the binding radius and the two tolerance regimes confirm statements the report already makes. The corrected τ constant of Section 7 and the unoccupied second branch of `prop: rdfo` in Section 5 are findings about the report rather than about the subsection.

10 Discussion

The stall is a property of the radius, not of the model. Every run in this report uses the exact Hessian. The model sees the escape direction perfectly and reports $\lambda_{\min} = -\varepsilon$ at the saddle. The method still takes 6362 iterations to move a distance $\frac{1}{2}$ at $\bar{\mu}\varepsilon = 10^{-3}$. (11) says why: the radius is exactly $\bar{\mu}\|g_k\|$ on every iteration, and near the saddle $\|g_k\| \approx \varepsilon|y_k|$, so the step the rule permits shrinks with the distance the run is trying to cover. The subproblem solver is not the constraint and neither is the model.

Two uses of second-order information, with different effects. The radius measure and the stopping test are both second-order and they act on different things. Replacing $\|g_k\|$ by τ_k in the radius shortens the escape by a factor of 8.6 at $y_0 = 10^{-3}$ and 18.0 at 10^{-6} , and Section 3 now predicts both to under half a per cent. The stopping test does something else: Table 5 shows that at `tol` = 10^{-5} thirty-six of forty cells stop at the saddle and report success, and lowering the tolerance removes all of it. Neither substitutes for the other and they are reported separately throughout.

The entry condition is where R-grad and R-DFO part. R-grad satisfies as: RCA at $k_0 = 0$ with no condition, so the stall is unconditional. R-DFO needs the run to enter the band first, and Table 4 shows k_0 ranging from 0 to 22 and three cells never entering. Where entry happens the stall follows, with a bound loose by a uniform factor of about two because γ_3 is a worst case. Where it does not, the behaviour on this family is immediate escape rather than the convergence to the saddle the proposition's second branch describes. That branch remains unexercised.

What the section can now claim. Three statements are supported by measurement rather than by argument. The growth envelope of `prop: escape` is never crossed, on any run, at any ε . The hypothesis as: RCA holds with equality on this family, so the bound is not resting on slack. And the bound is tight to 1.4 per cent over four decades of $\bar{\mu}\varepsilon$, degrading only where $\ln(1+t) \approx t$ stops holding. A bound that is one-sided, never violated and tight to a few per cent is worth stating as such.

What remains open. The second branch of `prop: rdfo` is not occupied on this scan and we do not know whether it is occupied anywhere on this family. The residual 0.1 to 0.3 per cent in

Section 7 has two candidate explanations and neither was isolated. Both are recorded rather than resolved.